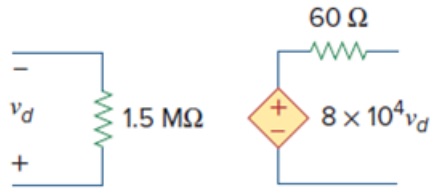


Problem

The equivalent model of a certain op amp is shown in Fig. 5.43. Determine:

- (a) the input resistance
- (b) the output resistance
- (c) the voltage gain in dB

Figure. 5.43:



Step-by-step solution

Step 1 of 3

(a)

$$R_{in} = 1.5 \text{ M}\Omega$$

Step 2 of 3

(b)

$$R_{out} = 60 \Omega$$

Step 3 of 3

(c)

$$A = 8 \times 10^4$$

Therefore,

$$\text{Gain in dB} = 20 \log \left(\frac{V_{out}}{V_{in}} \right)$$

$$A_{\alpha\beta} = 20 \log(8 \times 10^4)$$

$$A_{\alpha\beta} = 98.06 \text{ dB}$$